Entity-Token World Models and Differentiable Predictive Control for Building-Agnostic HVAC Control

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Abstract

A single controller for the Heating, Ventilation and Air Conditioning (HVAC) systems of buildings it has never seen has to cope with a different number of zones, air handlers and thermal properties in every building, as well as with climates ranging from subarctic winters to tropical summers. This work addresses the Smart Buildings Challenge with a policy that learns its own control logic, without deep reinforcement learning and without imitating an existing controller. Every zone, air handler, hot-water system and the building as a whole is represented as a token, and a transformer whose attention is biased only by the structural relation between two tokens is used both as a learned world model of the building and as the control policy, thereby making both networks independent of the building size. The policy is trained by Differentiable Predictive Control (DPC), that is, by backpropagating the competition reward, rebuilt from the predicted temperatures and power, through closed-loop rollouts of up to 24 hours inside the world model, while the world model is refined in successive on-policy rounds. On the 30 official scenarios evaluated over 14 days, the final policy reduces the mean absolute step reward from 0.0591 for the shipped PID baseline to 0.0106, and it reaches 0.013 on the Kaggle public leaderboard. The experiments further indicate that the calibration of the world model on the comfort term matters more than its average temperature error, and that a long planning horizon becomes beneficial only once the world model is accurate enough.

1 Introduction

Buildings account for a large share of energy use and carbon emissions, and their HVAC systems are still mostly operated by rule-based logic tuned by hand for each building [4, 13]. Model predictive control and reinforcement learning have repeatedly shown large savings on individual buildings, but both require a model or a training process specific to the building at hand, and this cost has so far prevented their deployment at scale [11]. The Smart Buildings Challenge [7], built on the Smart Buildings Control Suite simulator [6], makes this obstacle explicit by asking for one policy that controls buildings never seen during development, with different floor plans, numbers of zones and air handlers, envelope materials and weather, and that is scored on a reward combining occupant comfort, energy cost and carbon emissions.

The approach presented here rests on two choices. First, the building is described as a set of entities (zones, air handlers, the hot-water system and a global token for weather and time), and the same transformer architecture, with attention biased by the structural relation between entities, is used for a

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world model of the building and for the policy, so that both apply to any building without modification. Second, the policy is not trained by trial and error on the simulator, nor by imitation of a baseline controller, but by DPC [5]: the competition reward is rebuilt from the world-model predictions and maximized by backpropagation through closed-loop rollouts, thereby letting the policy derive its own control logic from the physics captured by the world model. The main contributions of this work are the following:

- an entity-token world model with long thermal memory, trained on procedurally generated buildings and refined in on-policy rounds, which reaches a closed-loop temperature error of 0.82 K after 24 hours on unseen buildings;
- a DPC training scheme with a 24-hour horizon, a command filter inside the rollouts and a comfort weighting of the reward, which yields a policy with a local score of 0.0106 and a Kaggle score of 0.013, starting from 0.0591 for the shipped baseline;
- an experimental analysis showing that the comfort calibration of the world model and its accuracy over long horizons, rather than its average short-term error, determine the quality of the resulting policy.

2 Related Works

The Smart Buildings Challenge [7] is the first benchmark in which a single HVAC policy is evaluated on unseen buildings generated at scale, using the simulator of the Smart Buildings Control Suite [6]; earlier benchmarks such as BOPTEST [2] provide detailed but individual test cases. Model Predictive Control (MPC) is the reference for advanced building control [4], although its dependence on a model identified for every building limits its scalability. Deep reinforcement learning has been applied to HVAC control since [14], and the review in [13] identifies sample efficiency, safety during training and the transfer to new buildings as its main obstacles; transfer learning across buildings is reviewed in [11], where most approaches still fine-tune a model on the target building.

Differentiable control combines the two families. Differentiable MPC [1] embeds an optimization problem in a policy, and Gnu-RL [3] applies it to building HVAC control with a linear model; DPC [5] instead trains an explicit neural policy offline by differentiating a control objective through a model of the system, and it is the formulation adopted in this work. Learned world models [8, 9] allow policies to be trained in imagination, and the exploitation of model errors by the policy, discussed in [10], is the main failure mode encountered in the present experiments as well; short-horizon actor-critic methods such as SHAC [15] address long horizons with a learned value function, an option that was tested here without success. The entity-token architecture follows the transformer [12], and the attention bias depending on the relation between two nodes is inspired by Graphormer [16].

3 Methods

3.1 Problem Setting

At every 5-minute step, the policy observes the zone air temperatures and setpoints, the air-handler and hot-water sensors, the outdoor air temperature and the time, and it commands the damper and reheat valve of every zone, the run state, supply-air temperature, static pressure and recirculation of every air handler, and the run state, supply-water temperature and differential pressure of the hot-water system. The step reward is

$$r_t = w_p \, R_t - w_e \, \tilde{E}_t - w_c \, \tilde{C}_t, \tag{1}$$

where $R_t \leq 0$ is the productivity regret of the occupants, which grows linearly and then quadratically with the distance of each occupied zone from its comfort band, and $\tilde{E}_t$ and $\tilde{C}_t$ are the energy cost and carbon emissions of electricity and natural gas, normalized by their maximum rates. The score of a policy is the mean absolute step reward over all steps and scenarios, so that lower is better.

3.2 Entity-Token Architecture

The architecture shared by the world model and the policy is outlined in Figure 1.

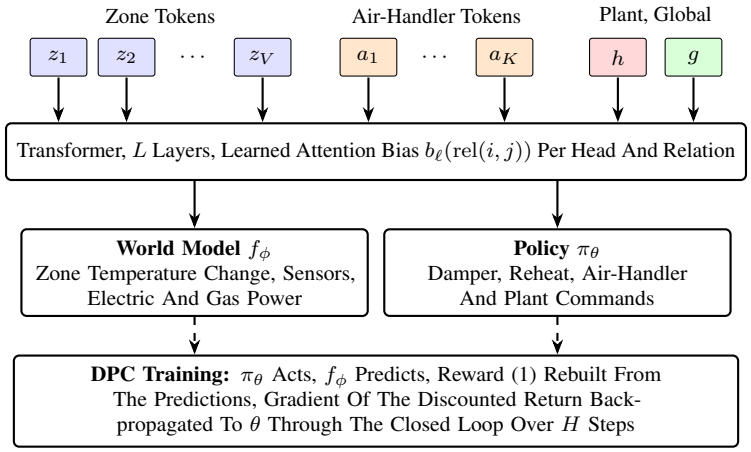


Figure 1: Entity-token architecture shared by the world model and the policy (same structure, separate weights), and the DPC training loop that connects them.

As can be observed in Figure 1, a building with V zones and K air handlers is represented by $V + K + 2$ tokens. Each zone token encodes its observations (temperature, setpoints, airflow, damper and reheat readbacks), the last three temperature changes, its previous commands and seven static descriptors extracted from the floor plan (logarithm of the floor area, exterior-wall fraction, exterior wall per square root of area, floor position, ground and top floor flags and number of zones served by the same air handler); air-handler and hot-water tokens encode their sensors and previous commands, and the global token encodes the outdoor conditions, the hour and day of the week and the occupancy schedule. Each token type has its own two-layer encoder and type embedding, after which pre-norm transformer layers [12] process all tokens jointly; the only structural information is a learned scalar per attention head added to the attention logit of each pair of tokens according to one of ten relations (for instance, two zones on the same air handler, a zone and its own air handler, any token and the hot-water system), in the spirit of Graphormer [16]. Since no parameter depends on V or K, the same weights apply to every building. The world model has 4 layers of width 128 (0.61 M parameters) and the policy 3 layers of width 64 (0.12 M parameters); on/off commands are trained with a straight-through estimator, while continuous commands pass through a hyperbolic tangent.

3.3 World Model

The world model f_ϕ predicts, for every zone, the change of air temperature over the next step and the next actuator readbacks, for every air handler and for the hot-water system their next sensor values, and the electric and gas power of the building, from which the reward (1) is rebuilt exactly, given the exogenous inputs (weather, occupancy, tariffs and carbon intensity) read from the episode. It is trained on multi-step unrolls of 4, 12 and 24 hours, in which its own predictions are fed back, with a loss combining the temperature error, weighted more for occupied zones near or outside their comfort band, the error on the comfort term of the reward and the errors on sensors and power. Since walls, slabs and interior mass store heat that is not observed, each zone token is augmented by a long memory made of exponential averages of its temperature over 1, 6 and 24 hours, of its reheat and damper commands over 6 and 24 hours and of the outdoor air over 6 and 24 hours, which the rollout updates from its own predictions.

The training data come from about 140 buildings produced by the official generator, with seeds disjoint from all evaluation sets, under six weather stations (Atlanta, Denver, Fairbanks, Minneapolis, Phoenix and Singapore). The first episodes were collected with an exploration policy switching every few hours between the shipped PID controller, its noisy version, randomly held setpoints and extreme plant states; every following round collects one 14-day episode per building with the current DPC policy, mostly as is and partly with action noise, and the world model is fine-tuned with half of each batch drawn from these on-policy episodes, thereby correcting it precisely in the states that the policy visits. The world-model versions discussed below differ in these rounds: v10 introduced the long memory and 24-hour unrolls, v11 added an on-policy round of the first filtered policy and v12 an

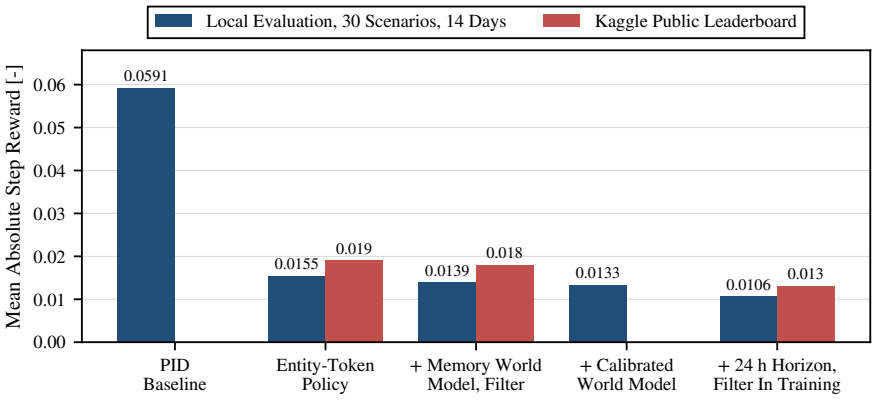


Figure 2: Mean absolute step reward of the shipped PID baseline and of the successive stages of the method, on the 30 official scenarios over 14 days and on the Kaggle public leaderboard.

on-policy round in which a quarter of the segments held the supply-water temperature, supply-air temperature and static pressure at random levels.

3.4 Differentiable Predictive Control

From start states drawn from the collected episodes, the policy π_θ acts, the world model predicts the next state and the reward is rebuilt from the prediction, so that the discounted return

$$J(\theta) = \mathbb{E}\Big[\sum_{k=0}^{H-1} \gamma^k \, \hat{r}_{t+k}\Big], \qquad \hat{r}_t = \kappa \, w_p \hat{R}_t - w_e \hat{\tilde{E}}_t - w_c \hat{\tilde{C}}_t, \tag{2}$$

is maximized by backpropagation through the whole closed loop, with $\gamma = 0.995$. The factor $\kappa = 1.8$ on the comfort term compensates for the tendency of the policy to sacrifice comfort for energy within the world model; the gradient carried by the predicted state from one step to the previous one is multiplied by $\lambda = 0.99$, so that credit assigned h steps later is weighted by λ^h and the backpropagation over hundreds of steps remains stable. The horizon H grows from 12 to 24 hours during training, which is long enough for the policy to see the effect of preheating before occupancy and of letting the building cool down at night. Since the policy reacts to world-model errors with fast oscillating commands, which the real plant does not reward, every continuous command is filtered as $\tilde{u}_t = \alpha u_t + (1-\alpha)\tilde{u}_{t-1}$ with $\alpha = 0.3$, both inside the training rollouts and at deployment. Each policy is trained for 400 iterations of 24 parallel rollouts with AdamW and a cosine schedule, starting from an earlier DPC policy; the start states exclude the generated buildings larger than the official size range, and the gradient steps whose loss is not finite are skipped.

4 Results

4.1 Evaluation Protocol

Policies are evaluated in the simulator on the 30 official training scenarios (six buildings under five weather stations), none of which is used for training, over 14 days (4032 steps), so that every episode includes a weekend and the following Monday start-up; earlier stages were evaluated over 7 days. The local 14-day score proved to be a reliable predictor of the hidden Kaggle evaluation: for the three policies submitted, the ratio between the Kaggle score and the local score was 1.23, 1.29 and 1.23, compared with 1.39 and 1.48 for the 7-day protocol.

4.2 Overall Performance

The progression of the method is summarized in Figure 2.

As can be observed in Figure 2, the first entity-token policy trained by DPC already reduces the score of the PID baseline from 0.0591 to 0.0155, that is, by 74%, and every following stage brings a further

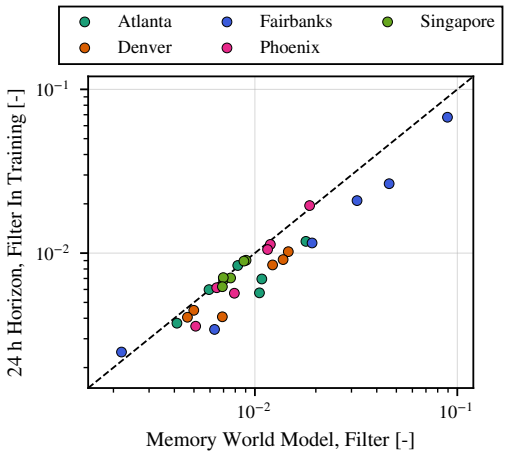


Figure 3: Score of each official scenario over 14 days, for the filtered policy on world model v10 (horizontal axis) and the final policy (vertical axis); points below the diagonal are improvements.

Table 1: World-model diagnostics after 24 hours (temperature mean absolute error and bias, ratio of predicted to true comfort and energy terms) and 14-day score of the policies trained on each model.

World Model	Replay MAE [K]	Closed-Loop MAE [K]	Closed-Loop Bias [K]	Comfort Pred./True	Energy Pred./True	Policy (Horizon) 14-Day Score [-]
v8	1.38	1.13	$+0.43$	0.96	0.95	Entity-Token (6–12 h): 0.0155
v9	1.50	1.35	$+1.07$	0.92	1.09	–
v10	0.87	0.87	-0.18	1.22	1.22	Filter (6–12 h): 0.0139
v11	0.89	0.82	$+0.06$	1.00	1.10	Filter (6–12 h): 0.0133 Filter In Training (24 h): **0.0106**
v12	0.78	0.79	-0.01	1.07	1.20	Filter In Training (24 h): 0.0121

reduction, down to 0.0106 for the final policy, which corresponds to 0.013 on the Kaggle public leaderboard, against 0.019 for the first entity-token policy submitted. The largest single gain comes from the 24-hour horizon trained with the command filter in the loop (from 0.0133 to 0.0106), whose effect on the individual scenarios is shown in Figure 3.

As can be observed in Figure 3, the final policy is better in 23 of the 30 scenarios, with the largest reductions in the cold climates (Fairbanks 1 from 0.089 to 0.068, Fairbanks 8 from 0.046 to 0.027), while in the remaining scenarios the differences are within a few percent. Figure 4 shows how this is achieved during the first week of Fairbanks 1, when the outdoor air drops below $-35\,^{\circ}\mathrm{C}$: the final policy delivers more heat in the second half of each working day, thereby keeping the coldest zones closer to the heating setpoint, at the price of an 11% higher gas consumption, and its mean step reward over the week improves from -0.0650 to -0.0548. Over the 30 scenarios the comfort term is more than halved (from 0.0065 to 0.0026), whereas the energy and carbon terms grow only slightly (from 0.0041 to 0.0043 and from 0.0033 to 0.0037, respectively).

4.3 World-Model Quality and Policy Quality

The world models were diagnosed on 45 clean episodes driven by a fixed policy, 30 on the official scenarios and 15 on unseen benchmark buildings, from 1080 start states, both by replaying the logged actions and in closed loop with the policy acting inside the model. Table 1 relates these diagnostics to the policies trained on each model.

As reported in Table 1, the long memory of v10 reduced the 24-hour error by about 35% with respect to v9, and the on-policy round of v11 removed both the temperature bias and the overestimation of discomfort (from 1.22 to 1.00). World model v12 has the lowest error of all, yet the same training recipe yields a worse policy on it than on v11 (0.0121 against 0.0106), better in only 7 of the 30 scenarios; v12 overestimates the comfort term by 7% and the energy term by 20%, and its policy

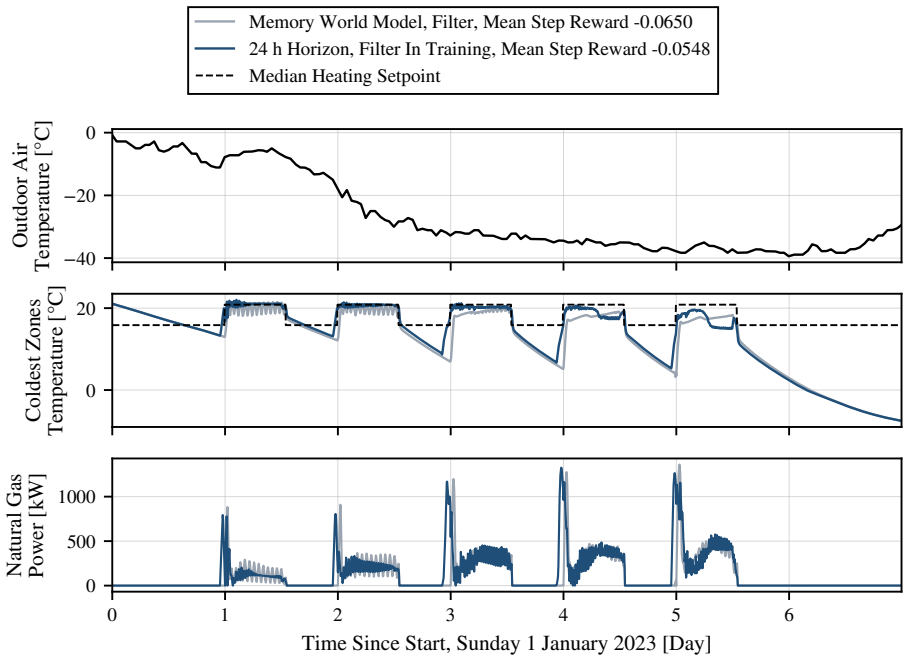


Figure 4: First week of Fairbanks 1 (official building 1): outdoor air temperature, 10th percentile of the zone temperatures and natural gas power of the filtered policy on world model v10 and of the final policy.

Table 2: Effect of the main design choices on the official scenarios (7 or 14 days, as indicated); every row compares a variant with the reference trained in the same conditions.

Design Choice	World Model, Days	Reference [-]	Variant [-]	Change
Zone PI controllers supervised by the policy	v8, 7	0.0151	0.0179–0.0302	worse
No comfort weighting ($\kappa = 1$)	v8, 7	0.0151	0.0137	better
No comfort weighting ($\kappa = 1$)	v10, 7	0.0156	0.0299	worse
Smoothness penalty on commands	v10, 7	0.0156	0.0178–0.0229	worse
Command filter at deployment ($\alpha = 0.3$)	v10, 7	0.0156	0.0122	better
Horizon 12–24 h instead of 6–12 h	v9, 7	0.0195	0.0328	worse
Horizon 12–24 h, filter in training	v11, 14	0.0133	0.0106	better
Filter in training, horizon 6–12 h	v11, 14	0.0133	0.0182	worse
Policy memory and holiday detection	v12, 14	0.0121	0.0117	better

spends more energy than needed, especially in the warm climates. This can primarily be attributed to the fact that the DPC gradient follows the predicted reward rather than the predicted temperature, so that a systematic bias on the reward terms steers the policy, whereas a random temperature error largely averages out.

4.4 Design Choices

Table 2 collects the main design choices, each compared with the reference trained on the same world model.

As reported in Table 2, letting the policy act directly on the actuators is better than letting it supervise zone-level PI controllers, which constrain the control logic to a predefined structure. The comfort weighting κ is not needed on v8, but it becomes essential on v10, where its removal doubles the score, and it has therefore been kept for all later policies; similarly, the long horizon fails on the inaccurate v9, where the policy exploits errors that accumulate over 24 hours, and pays off only on v11. Penalizing the variation of the commands did not remove the oscillations caused by world-model errors and worsened the score, whereas filtering the commands did; the filter, however, must be

combined with the long horizon when it is applied during training, since with short rollouts it slows down the response more than it helps. Finally, giving the policy the same long memory as the world model, together with the detection of non-working days from the setpoint band, improves the score by about 3%, mostly in the cold climates (Fairbanks 1 from 0.066 to 0.058). A value function bootstrapping the DPC return beyond the horizon, in the spirit of SHAC [15], was also tested in two forms, but it remained poorly calibrated and degraded the policy, and it was therefore abandoned.

5 Conclusion

This work has shown that a single policy for unseen buildings can be obtained without deep reinforcement learning and without imitation, by representing the building as a set of entities, learning a world model of its physics and differentiating the competition reward through closed-loop rollouts of that model. The resulting policy reduces the score of the shipped PID baseline by 82% on the official scenarios and reaches 0.013 on the Kaggle public leaderboard. The experiments indicate that the main driver of performance is the world model, and in particular its calibration on the reward terms and its accuracy over the planning horizon, since each improvement of the world model made a longer horizon or a weaker regularization usable. The current results rest on a single training seed per configuration, and the remaining gap to the best entries of the leaderboard is concentrated in the coldest climates. The next steps are the measurement of the variance between training seeds, ensembles of world models to make the policy robust to the errors of any single model, and planning at deployment, in which the actions of the policy are refined online by a short search inside the world model.

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